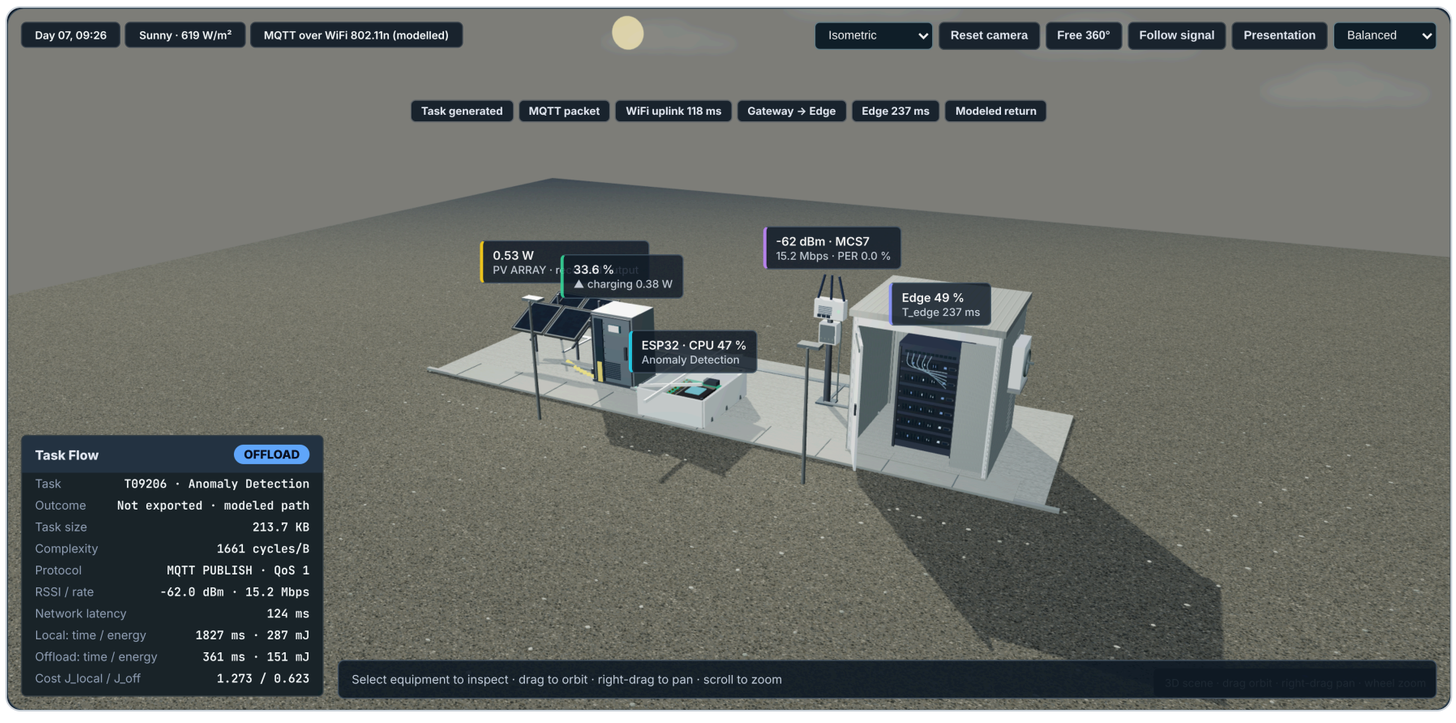
19CSE362 · Fundamentals of Soft Computing

Methodology and Results

Explainable ANN–ANFIS hybrid for renewable-aware task offloading

1 Digital twin and dataset



The project's own 3D twin replaying task T09206 (day 7, 09:26, sunny). Running it locally would take 1.8 s and 287 mJ; offloading takes 0.36 s and 151 mJ, so it is offloaded.

Solar panel
power = 1 W × sunlight ÷ 1000 × 0.85

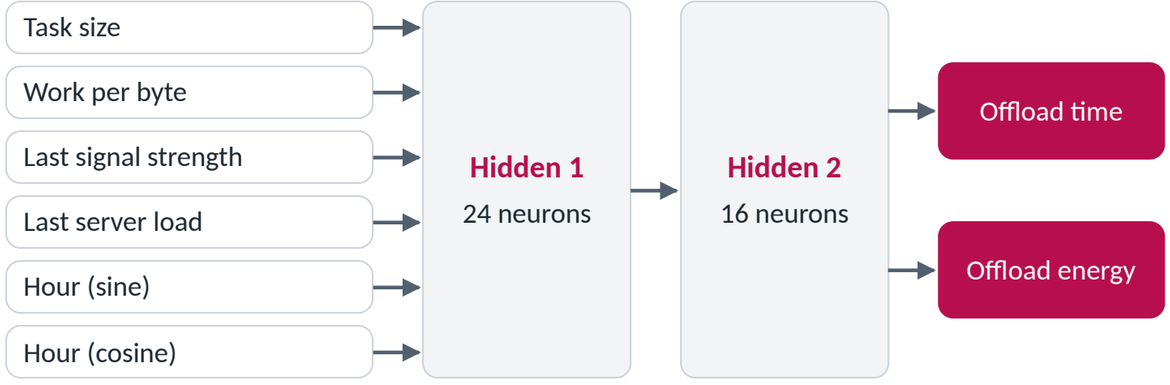
Battery (12 Wh)
goes up with solar, down with power used

ESP32 (240 MHz)
work = data size × work per byte; time = work ÷ free CPU speed

WiFi and edge server
weaker signal → slower link and retries; a busy server is slower

Dataset
7 days → 10,080 tasks × 38 features, each labelled with the cheaper option

2 ANN predictor

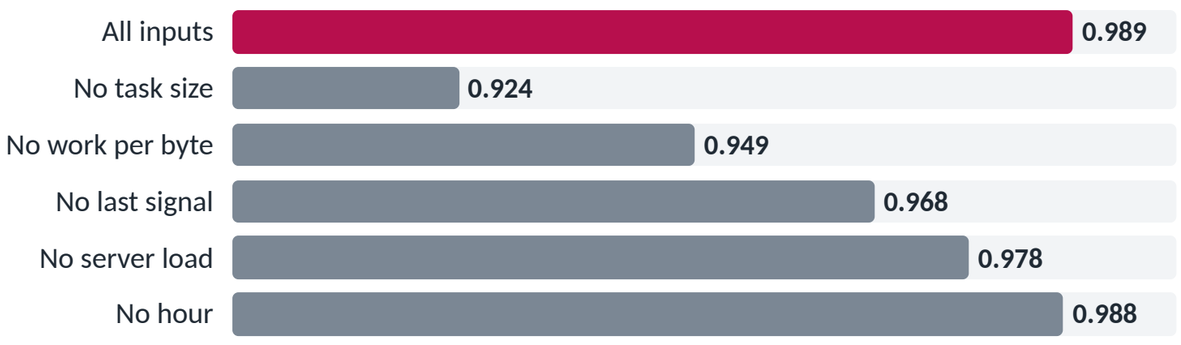


0.989 accuracy for time (R^2)

0.987 accuracy for energy (R^2)

≈ 5 % median error

Ablation: remove one input and retrain (time accuracy, R^2)



Bars start at 0.90. Task size and work per byte matter most.

3 ANFIS: 36 learned rules

1 · Fuzzify how Low, Medium or High each input is → **2 · Rules** multiply the matches (AND) → **3 · Share** each rule's share of the total → **4 · Output** a straight-line formula per rule → **5 · Add up** offload score; above 0.5 → offload

Inputs: battery level (Low, Medium, High) · energy ratio from the ANN (Offload cheaper, Similar, Local cheaper) · time gain from the ANN (Small, Large) · server load (Light, Busy) → 3 × 3 × 2 × 2 = **36 rules**. Trained by least squares plus gradient descent against a soft "how much cheaper" target.

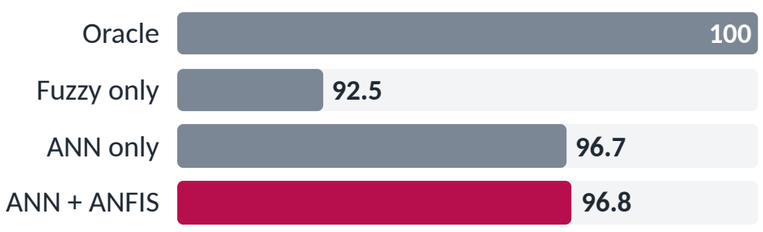


The learned membership curves, from the project's dashboard.

Task T09206: the ANN predicts 353 ms and 149 mJ (real 361 ms, 151 mJ). The strongest rule is *Battery Medium AND Offload cheaper AND Small gain AND Busy server*, and the score of 0.9965 means **OFFLOAD**.

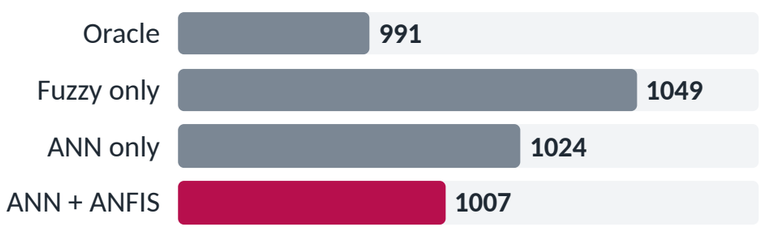
4 Results on 11 unseen weeks

Agreement with the oracle (%)



Bars run from 90 to 100 %. Higher is better.

Task energy per week (J)



Bars run from 950 to 1075 J. Lower is better.

vs Fuzzy only: better in all 11 weeks on accuracy, energy and extra cost (Wilcoxon p ≈ 0.001).

vs ANN only: the same accuracy, but less energy in all 11 weeks (p ≈ 0.001), and it explains every decision.

Next: calibrate on real hardware, add a cloud tier, run ANFIS on the ESP32.